

Course Title	Linear Algebra		
Course Code	MC215	Credit Structure	3-1-0-4 (L-T-P-C)
Category	Core	Semester	Winter Semester (AY 25-26)
Program	B.Tech-MnC		
Prerequisites	NA		
Course Objectives/ Brief Course Description	Vectors are an essential idea to understand numerous natural phenomena. Its applications start right from mechanics and electrostatics and magnetostatics. These applications use equations involving vectors. Such ideas are abstracted to study solutions of a system of linear equations in several variables.		
Evaluation/ Grading Policy	In-Sem I: 30%, In-Sem II: 30%, End-Sem 40%, Relative grading		
Course Materials/ References	1. <i>Linear Algebra</i> , S. H. Friedberg, A. J. Insel and L. E. Spence 2. <i>Linear Algebra Done Right</i> , Sheldon Axler 3. <i>Linear Algebra</i> , K. Hoffman and R. Kunze 4. <i>Linear Algebra: A Geometric Approach</i> , S. Kumaresan		
Detailed Course Content	APPENDIX		

Course Outcome:

- CO1: Interpret existence and uniqueness of solutions geometrically. Understand algebraic and geometric representations of vectors in $\mathbb{R}^n$ and their operations, including addition, scalar multiplication and dot product. Solve systems of linear equations using Gauss- Jordan elimination to reduce to echelon form. Solve systems of linear equations using the inverse of the coefficient matrix when possible. Understand the nature of the solutions of systems of linear equations. [PO1, PO2, PO12]
- CO2: Determine a basis and the dimension of a finite-dimensional space. Discuss spanning sets for vectors in $\mathbb{R}^n$. Discuss linear independence for vectors in $\mathbb{R}^n$. Define the dimension of a vector space. Prove all bases have the same number of elements. Prove elementary theorems concerning rank of a matrix. State and prove the rank-nullity theorem. Demonstrate matrix representation of linear operators and understand the concepts through linear operators and matrices. [PO1, PO2, PO12]
- CO3: Find the eigenvalues and eigenvectors of a matrix. Define eigenvalues and eigenvectors geometrically. Use characteristic polynomials to compute eigenvalues and eigenvectors. Use

eigenspaces of matrices, when possible, to diagonalize a matrix. Perform diagonalization of matrices. Explain the significance of eigenvectors and eigenvalues. [PO1, PO2, PO12]

POs-COs Matrix:

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
x	x	x	x								x

APPENDIX: Detailed Course Content (Session-wise/ Module-wise)

Topic	Content	Number of Lectures
System of linear equations	System of linear equations, Elementary row operations, Row reduced echelon form	3
Vector spaces	Definition, Example, Basis and dimension, Subspace, Direct sums	7
Linear transformations	Definition, Example, Rank-nullity theorem, Algebra of linear transformations, Representation of linear transformations by matrices, Change of bases, Linear functionals and annihilators, Dual spaces.	10
Eigenvalues and eigenvectors	Definition, the characteristic polynomial, the Cayley- Hamilton Theorem, the minimal polynomial, algebraic and geometric multiplicities, Diagonalization, Jordan canonical form.	10
Inner product spaces	Definition, orthonormal bases, Gram-Schmidt process; Adjoint of operators, normal, unitary, and self-adjoint operators, spectral theorem for self-adjoint and normal operators	10